# GENERAL INSURANCE CAPITAL & RISK MODELLING

## Executive Actuarial Analytics Report

**A frequency-severity capital modelling case study using Python to quantify aggregate insurance losses, tail risk, stress scenarios and diversification.**

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# 1. Executive Summary

This case study develops a simplified one-year general insurance capital risk model using the public freMTPL2 French motor third-party liability dataset. The analysis combines policy-level claim frequency modelling, heavy-tailed severity modelling and Monte Carlo aggregate-loss analysis to distinguish expected claims cost from unexpected capital risk.

| Metric | Result |
|---|---|
| Expected annual loss | **£64.68m** |
| 99.5% Value at Risk (VaR) | **£82.63m** |
| 99.5% Tail Value at Risk (TVaR) | **£85.17m** |
| Unexpected loss at 99.5% | **£17.94m** |
| Capital uplift over expected loss | **27.74%** |
| Combined +10% stress VaR | **£98.97m** |
| Diversification benefit | **£0.72m (3.85%)** |
| Tail share of standalone capital | **96.23%** |

**Central finding: body and tail claims contribute approximately equally to expected loss, but large-loss tail claims account for 96.23% of standalone unexpected-loss capital.**

The results show why an insurer should not rely on expected loss alone when assessing financial resilience. Capital requirements are driven disproportionately by adverse tail outcomes, and the model is materially more sensitive to severity deterioration than the expected-loss split alone would suggest.

# 2. Business Question & Data

The management question is: how much capital should a general insurer hold against losses that may be materially worse than expected, what risks drive that requirement, and how does the position change under adverse scenarios?

## Dataset

The project uses the public freMTPL2 French motor third-party liability data. The frequency dataset contains 677,991 policy-level records, while the severity dataset contains 26,444 claim records. The observed severity sample totals approximately £59.91m of claim cost, with a mean claim of £2,265.51 and a maximum observed claim of £4.08m.

## Modelling workflow

- Data quality assessment and exploratory analysis.
- Poisson and Negative Binomial claim-frequency modelling.
- Lognormal and Gamma severity comparison, followed by Generalised Pareto tail modelling.
- Composite body-tail severity model with a £4.08m maximum modelling loss.
- Monte Carlo simulation of one-year aggregate losses.
- VaR, TVaR, stress testing, risk contribution and diversification analysis.

## Key model choices

The Negative Binomial GLM was retained for frequency because it improved fit relative to the Poisson baseline. For severity, a Lognormal body and Generalised Pareto tail were used because simple parametric distributions materially understated the extreme observed tail. The selected tail threshold was £8,156.01 (97.5th percentile), with the maximum modelling loss capped at the observed maximum of £4,075,400.56 to control unbounded heavy-tail behaviour.

## 3. Capital Results

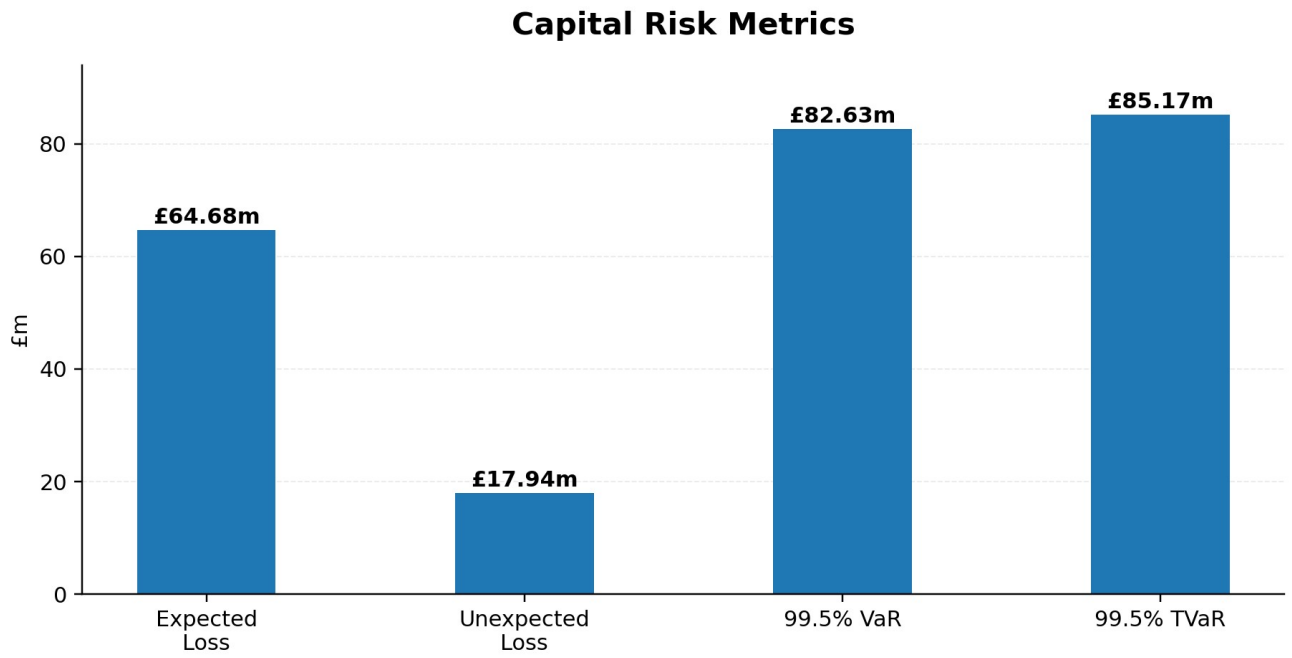



Figure 1. Core one-year capital risk metrics from the base simulation.

The model produces an expected annual loss of £64.68m and a 99.5% VaR of £82.63m. The difference, £17.94m, represents unexpected loss at the 99.5% confidence level and is 27.74% above expected loss. The 99.5% TVaR of £85.17m is higher than VaR because it measures the average loss in the most adverse 0.5% of simulated outcomes rather than the threshold alone.

**The 99.5% VaR is used as a risk metric in this case study. It should not be interpreted as a regulatory Solvency Capital Requirement.**

## 4. Stress Testing

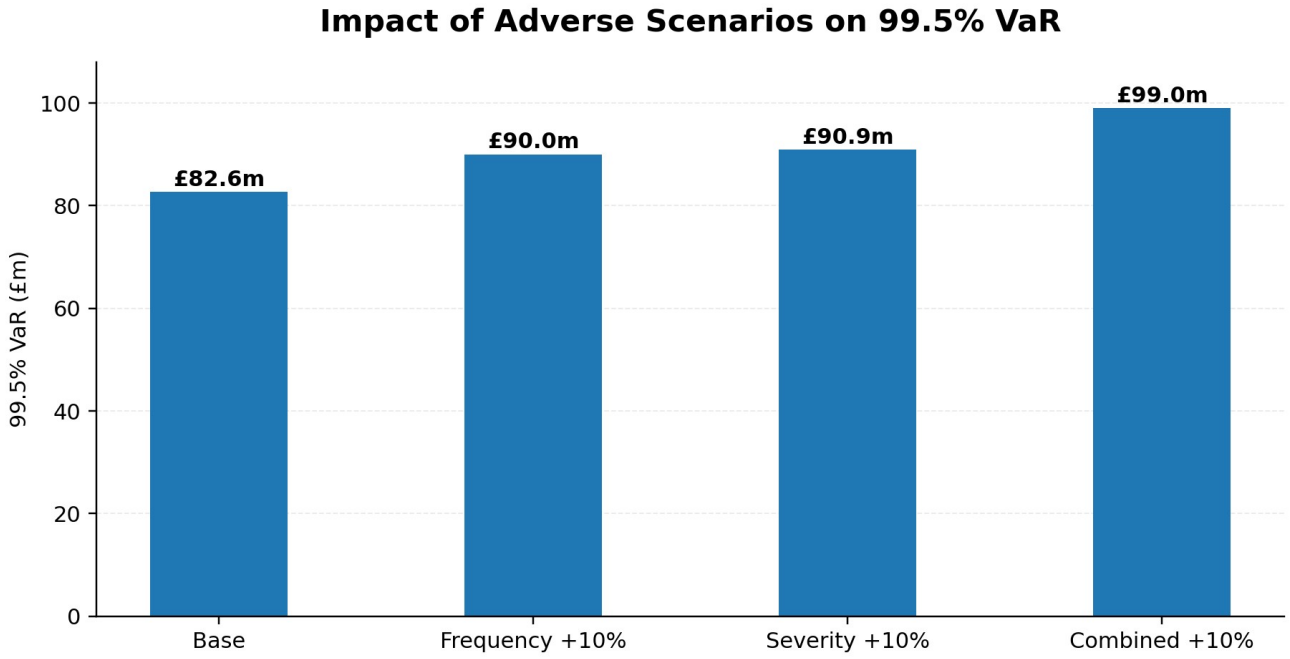



Figure 2. Effect of illustrative +10% frequency and severity stresses on 99.5% VaR.

A 10% frequency increase raises 99.5% VaR from £82.63m to approximately £89.97m, while a 10% severity increase raises it to £90.89m. Although the two individual stresses produce the same proportional increase in expected loss, severity deterioration has the larger effect on unexpected loss.

Applying both stresses together increases 99.5% VaR to £98.97m, approximately 19.8% above the base VaR. This demonstrates the value of considering adverse scenarios alongside central forecasts when assessing financial resilience.

## 5. Risk Contribution & Diversification

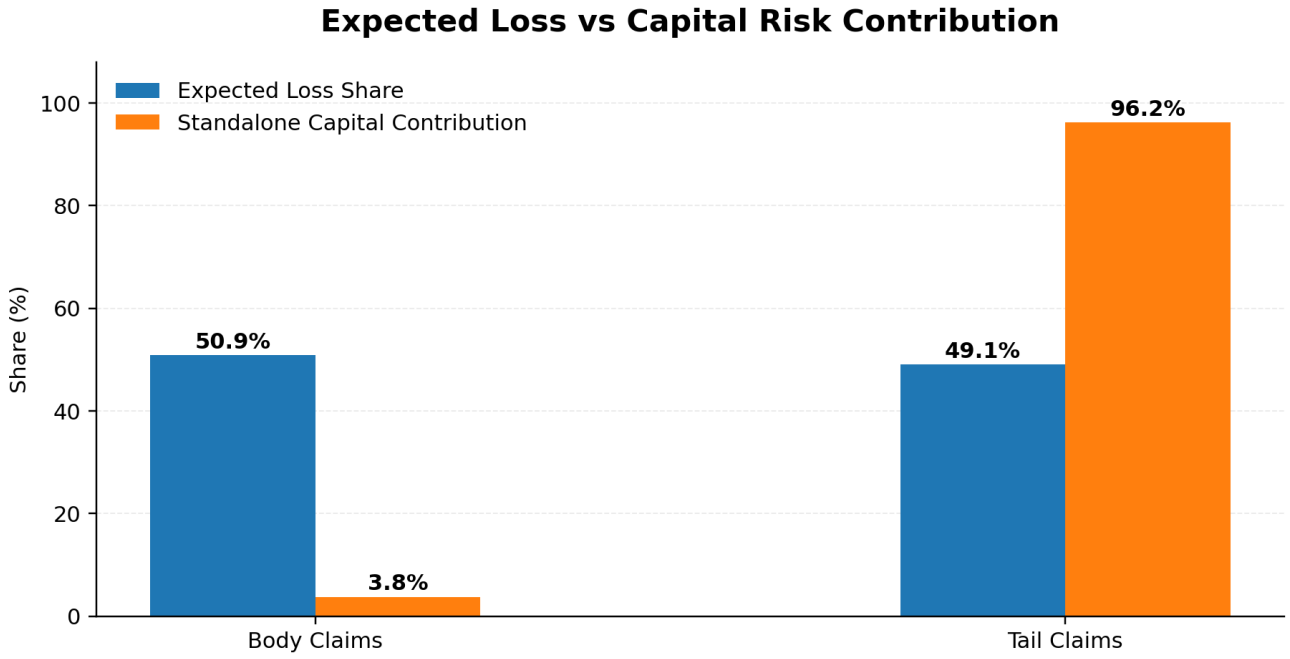



Figure 3. Expected-loss share compared with standalone unexpected-loss capital contribution.

Body claims contribute 50.9% of expected loss and tail claims 49.1%, yet their capital contributions are very different. Body claims account for only 3.77% of standalone unexpected-loss capital, while tail claims account for 96.23%.

The sum of standalone capital is £18.66m compared with diversified capital of £17.94m, producing a diversification benefit of approximately £0.72m, or 3.85%. Within this simplified framework, diversification reduces required capital modestly, while large-loss tail behaviour remains the dominant capital driver.

**Management implication: controlling and understanding large-loss severity is more important for capital resilience than its expected-loss share alone would suggest.**

## 6. Management Implications

- Expected loss and capital risk should be monitored separately: the claims that generate routine cost are not necessarily the claims that drive solvency pressure.
- Large-loss severity deserves focused monitoring because the tail dominates standalone capital contribution.
- Stress testing should complement base-model results, particularly where frequency and severity may deteriorate together.
- Diversification should be recognised, but not treated as a substitute for understanding the underlying tail risk.
- Capital decisions should be supported by sensitivity analysis around tail thresholds, severity limits and dependence assumptions.

## 7. Model Scope & Limitations

This is an actuarial analytics case study rather than a regulatory capital model. It considers insurance claim frequency and severity risk only. Market, counterparty/default, operational and other material insurer risks are outside scope. Frequency and severity are modelled independently, and body and tail components are treated independently in the diversification analysis.

The large-loss model is sensitive to the selected Generalised Pareto threshold and fitted parameters. A maximum modelling loss of £4.08m is applied to control the unbounded behaviour of the fitted heavy-tail distribution. The stress scenarios are illustrative management stresses rather than prescribed regulatory stresses, and historical claim experience may not fully represent future portfolio conditions.

### Conclusion

The case study demonstrates a complete frequency-severity capital modelling workflow from raw-data validation through to decision-focused capital analysis. Its main result is that tail severity, rather than expected-loss share, is the dominant driver of unexpected capital in this portfolio.

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